

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi

CS406: Mid-semester Examination
25th January 2023

Max. marks: 20

Time: 1 hr

Q 1. Find the real root of the equation $3x = \cos x + 1$, using Newton-Raphson method.

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Q 2. Solve the following system of equations using the Gauss-Jordan method

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$$\begin{aligned}x + y + z &= 1 \\4x + 3y - z &= 6 \\3x + 5y + 3z &= 4\end{aligned}$$

Q 3. Find the smallest positive root of $x^2 - 5x + 1 = 0$, correct to four decimal places

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Q 4. Test the consistency of the following system of equations

$$\begin{aligned}x + 10y - z &= 3 \\2x + 3y + 20z &= 7 \\9x + 22y + 79z &= 45.\end{aligned}$$

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Q 5. Using Lagrange interpolation, find the unique polynomial $P(x)$ of degree 2 or less such that $P(1) = 1$, $P(3) = 27$, $P(4) = 64$.

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